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# Trust or consequences? Causal effects of perceived risk and subjective norms on cloud technology adoption

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## ABSTRACT

Cloud computing has become a popular alternative information curation solution for organizations. As more corporate proprietary information is stored in the Cloud, concerns about Cloud information security have also increased. This study investigates the causal effect of perceived risk and subjective norms on users' trust intention to adopt Cloud technology. A partial least squares structural equation modeling (PLS-SEM) analysis was performed to assess latent variables and examine moderating effects to Cloud technology adoption. Our findings suggest that a user's perceived risk and subjective norms have a significant effect on their trust intention vis-a-vis Cloud adoption, which leads to their decision on whether to adopt the Cloud technology. While a user's attitudes do influence their intention to trust the Cloud, their attitude is not moderated by either perceived risk or subjective norms. On the other hand, a user's perceived risk of the Cloud environment predominately moderates their knowledge and perceived behavioral control, which results in their knowledge and perceived behavioral control not having a direct effect on their intention to trust the Cloud. Moreover, a user's subjective norms predicate their trust intention, which further moderates their knowledge and perceived behavioral control to trust and to adopt Cloud technology.

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## 1. Introduction

The National Institute of Standards and Technology (NIST), an agency under the aegis of the United States Department of Commerce, defines *Cloud computing* as follows:

“Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly

provisioned and released with minimal management effort or service provider interaction” (NIST SP 800-145, 2011).

The ubiquitous access provided by Cloud computing has undoubtedly afforded us the opportunity to conduct both personal and business matters with greater ease and efficiency. However, the expansion and growth of the Cloud computing environment has not been accompanied by an equivalent increase in security countermeasures, but has become a black box to end-users (Armbrust et al., 2010). Although corporate data can be made accessible from anywhere, users have very little control

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over the access, use or dissemination of information stored “in the Cloud.” Organizations cannot ignore the fact that once their corporate proprietary information is transferred over to a Cloud service provider, it can no longer be considered private and confidential. Even changing service providers after information has been uploaded does not restore the privacy or confidentiality of the information. Thus, it is unlikely for an organization to change its Cloud service provider once established. Cloud computing presents enormous challenges in terms of security, privacy, and trust for both Cloud providers and users (NIST SP 800-144, 2011).

In this study, we are interested in identifying the extent to which the intention of corporate users’ trust the Cloud influences their decisions to adopt this technology. Moreover, we seek to determine causal effects of perceived risk and subjective norms on corporate users’ trust intention of the Cloud, and how these factors moderate users’ knowledge, attitudes, and perceived behavioral control of their information in the Cloud. Thus, our overarching research question is: *To what extent are users’ decision to adopt Cloud technology directly impacted, and indirectly moderated, by perceived risk and subjective norms?*

We begin this paper with a general view of the Cloud computing environment as a type of open system. We critique and compare models – specifically the technology adoption model (TAM) and the theory of planned behavior (TPB) – to assess which best explains or predicts a user’s intention to trust. In the subsequent sections, we discuss our research model and hypotheses. With data collected from corporate IT managers and decision makers, we discuss the factor loadings for our research model, and apply partial least squares structural equation modeling (PLS-SEM) to examine not only factors that directly affect corporate users’ trust intention and behavioral outcome in adopting Cloud technology, but also factors that moderate users’ trust intention. The paper concludes with a discussion of the contributions to research and practice, limitations, and potential future research direction(s).

## 2. Adoption of cloud computing services

“The Cloud” functions much like an “open system” (Chau and Tam, 2000, p. 231). This “open system” offers three basic types of service: Software as a Service (SaaS), Platform as a Service (PaaS), and Infrastructure as a Service (IaaS) (NIST SP 800-145, 2011). These service models establish a level of control over the corresponding applications and data, and can be deployed in any of four basic deployment models: private, community, public, and hybrid (NIST SP 800-145, 2011). These deployment models establish a community of users as having access to that particular shared information, interest, and networked resources. Through the increased interoperability and portability of functions offered by this particular open system, adoption of Cloud computing has resulted in fundamental, transformational changes in organizational behavior. In particular, adopting Cloud services enables organizational users to respond with greater effectiveness and efficiency to the demand(s) of their client(s) (Marston et al., 2011). Likewise, the Cloud has the potential to enhance collaboration, agility, scaling, and availability, while providing opportunities for operational cost reduction through optimized and efficient computing (Armbrust et al.,

2010). Cloud computing is considered a highly viable alternative to a corporate information technology (IT) infrastructure for internal information organization and management.

Despite these potential benefits, there are fundamental concerns about the security of information stored in the Cloud, including risk of data loss, information control/oversight, and compliance of information management services. Indeed, security concerns have become the most frequent objection by organizational users to the adoption of Cloud computing (Oliveira et al., 2014). An organization’s decision to adopt the Cloud can be viewed as a decision made by the organization as a whole. The organizational decision to adopt the Cloud as its information infrastructure may be influenced by a variety of economic factors – including the potential for increased economic value (e.g., realization of the technology benefits, improved user experience) and market value (i.e., readiness of the market, vendor’s reputation) (Son et al., 2014). Specifically, corporate decisions to adopt the Cloud are based on firm-specific, resource-dependent, vendor-specific factors. There are also a number of logistical factors – internal and external – that may influence an organizational decision to adopt Cloud infrastructure. Internal factors include the complexity of the organization’s existing IT infrastructure and its compatibility with the Cloud infrastructure, as well as the organization’s structure and level of support among employees and management for adoption. External influencers include the readiness of the Cloud technology itself, competitive pressure/competitive advantage, and level of regulatory support (Oliveira et al., 2014).

However, an organization’s decision to adopt the Cloud can also be viewed as a decision made by individuals on whether they trust the Cloud for information storage, and which Cloud provider to trust. Trust is critical in the context of an organization that delegates an individual or a group to manage a project or service with a high-risk profile, such as managing the organization’s IT infrastructure. In the context of Cloud technology adoption, intention to trust is even more complex and critical to the ultimate decision to adopt technology. Users’ intention to trust is often influenced by an actual demonstration of security management by the Cloud service provider. Tang and Liu (2015) examined factors that influence trust intention, and proposed a model that includes objective evaluation of the trustworthiness of Cloud service providers in four areas of consideration – Function, Auditability, Governability, and Interoperability (FAGI). Moreover, users’ intention to trust can also be influenced by their own knowledge, attitude, and how much control they have over the systems. Lacohee et al. (2006) conducted focus group workshops to examine users’ attitudes, perceived risk and online trust in the adoption of Cloud technology. Users who conduct a basic risk assessment are more likely to increase their trust intention, and thus more likely to adopt the Cloud system. Shiau and Chau (2016) proposed a unified model for Cloud adoption that includes the technology acceptance model, service quality, innovation diffusion theory, theory of planned behavior, motivation model, and self efficacy. Overall, an organization’s decision to adopt Cloud technology depends on the attitudes, knowledge, perceived behavioral control, and intention to trust of the decision-makers themselves. These factors – along with perceived risk and subjective norms (e.g., influence from friends or organizational seniors) – form the core of our investigation.

In the section below, we compare and critique two theoretical frameworks – Technology Adoption Models (TAM) and the Theory of Planned Behavior (TPB) – that provide insight into how these different factors influence a user's intention to trust the adoption of Cloud computing services.

### 2.1. An aggregated critique of the technology adoption model

Davis (1989) proposed a technology acceptance model which asserted that two factors were of particular importance in the decision to adopt any particular technology: the *perceived usefulness* and the *perceived ease of use* of the technology. These two factors shape the user's beliefs and behavioral intention towards the technology, which can influence the behavioral outcome of adopting the technology. In addition to these two factors, Venkatesh and Davis (2000) asserted that social influence (i.e., subjective norms, voluntariness, and personal image), facilitating conditions (i.e., environmental factors) and cognitive instrumental processes (i.e., job relevance, output quality, and result demonstrability) influence users' intention to adopt technology. They also posited that users' expectations on performance and effort also affect the adoption decision. Venkatesh et al. (2003) further proposed a unified TAM theory (TAM2), which suggested that users' behavioral intention to use and adopt technology is significantly influenced by four research constructs: performance expectancy, effort expectancy, social influence and facilitating conditions.

Although TAM can be considered a fairly straightforward approach to understanding users' technology adoption behavior, Mathieson (1991) raised several significant criticisms of Davis's (1989) initial TAM framework. First, TAM makes some fundamental assumptions about users' opinions: (1) users' perceptions of ease of use and usefulness do not vary in different circumstances, and (2) the only relevant factors influencing technology adoption behavior are ease of use and usefulness. In other words, TAM effectively leaves out factors *other than* ease of use and usefulness that might influence a user's decision to adopt a particular technology. Second, Mathieson (1991) noted that TAM lacks consideration of social norms. TAM "does not explicitly include any social variables," whereas Ajzen's (1985, 1987) Theory of Planned Behavior (TPB) does. Third, Davis's (1989) TAM omits consideration of the user's perceived control as a factor influencing the user's decision to adopt technology (Mathieson, 1991, p. 179). Legris et al. (2003) also criticized TAM, asserting that broader perspectives from human and social change process should be integrated into TAM in order to understand and predict users' technology adoption decisions and behaviors.

Researchers have investigated the influence of social processes on perceived usefulness in technology adoption. For example, Schepers and Wetzels (2007) further adopted a statistical meta-analysis to examine the moderating effects of subjective norms, and found additional insights – including individual-related factors (i.e., type of respondents), technology-related factors (i.e., type of technology) and contingent factors (i.e., culture) – that has indirect influence on perceived usefulness and behavioral intention to use technology. Venkatesh and Goyal (2010) also noted that corporate users' perceived risk will likely affect their decision to adopt a particular product

or service. Shropshire et al. (2015) found that elements of personality – specifically, agreeableness and conscientiousness – have a moderating effect on users' behavioral intention to adopt security software.

Technology adoption is context specific. Straub et al. (1997) noted that TAM may not predict technology use across all cultures. When Hong and Tam (2006) examined innovation adoption in the context of multipurpose information appliances, they empirically tested this adoption model as applied to mobile data services, and concluded that users' adoption decisions are not only context specific, but are also dependent on the nature of the target technology and its usage context.

Users' rational decisions are significantly influenced by their intention to trust; however, investigation of trust was not included in the original TAM. Gefen et al. (2003) identified the role of trust specifically in e-Commerce adoption. Typically, online shoppers' perceived ease of use of e-Commerce will influence their trust intention, which will in turn influence their perception of the usefulness of the e-Commerce site and impact their intention to use e-Commerce. Similarly, de Guinea and Markus (2009) pointed out that, in addition to rational decision-making, emotional/ subconscious responses also influence a user's behavior when adopting a technology. A core problem in IS research is that users' actions, most of the time, do not require conscious behavioral intention (i.e., unplanned and unreasoned action). Accordingly, de Guinea and Markus (2009) asserted that it is critical to reexamine TAM from this perspective.

### 2.2. An argument for theory of planned behavior

Ajzen (1985, 1991) posited that a user's attempt to perform an action is often influenced by his or her beliefs about the perceived probabilities and consequences of success or failure. A user also tends to reference the opinions of others in making such decisions, and thus can be influenced by his or her normative beliefs. As a result, the user will attempt to perform an action only if: 1) s/he believes the advantages of success outweigh the disadvantages of failure; 2) s/he believes s/he will likely be successful in the attempt; and 3) the consensus is that s/he should do so. This latter factor (normative behavior) is informed not only by experiences, but also by the user's sense as to whether s/he has sufficient control over requisite internal and external factors (e.g., self vs. environmental factors). Ajzen (1985) thus posited three independent determinants of intentions: *attitude* toward the behavioral intention, *subjective norms* influencing the behavioral intention, and the degree of *perceived behavioral control* (i.e., the self-assessed ability to perform a behavioral outcome). A main assumption of TPB is that intentions capture those motivational factors that can influence the user's behavior (Ajzen, 2005), and thus intentions can be used to predict the effort an individual is likely to expend in performing – or, more precisely, attempting to perform – that behavior (Ajzen, 1991). Related to the foregoing is the additional assumption that the stronger the intention is to engage in a behavior, the more likely the performance of that behavior will be (whether attempted or achieved). However, Ajzen (1991) pointed out that intention manifests in actual behavior only if the individual is free to choose whether or not to perform

the behavior. Independent, non-motivational factors such as ability (i.e., availability of resources, time, money, and involvement of any necessary third parties) and opportunity may be a deterrent to actual performance of the intended behavior, irrespective of the strength of the individual's intention.

Taylor and Todd (1995) proposed an integrated TAM/TPB model, and suggested that *perceived usefulness* and *perceived ease of use* may have a direct effect on behavioral intention to adopt. Their empirical results demonstrated that *perceived usefulness* significantly influence users' attitudes towards the particular technology, and thus influence heavily their intention to adopt. In this regard, TPB subsumes these core concepts of the original TAM (i.e., perceived usefulness and ease of use) in the TPB core construct, *attitude*. Venkatesh et al. (2003) further noted that gender, age, users' experience and voluntariness of use also have *moderating* effects on the constructs of performance expectancy, effort expectancy, and social influence over users' intention to adopt.

In short, TPB takes into account the influence of *attitude*, *subjective norms* and *perceived behavioral control* on intention and behavioral outcome, and thus, we suggest, is better suited to explore users' intention to trust the Cloud technology adoption.

### 3. Theoretical frameworks

Our research model assumes in the first instance that, by design, the Cloud is both useful and easy to use. Thus, these two pillars of TAM are considered constant. Accordingly, our model specifically looks at the influence of users' perceptions on their intention to trust the Cloud and thereby influence behavioral

outcome (i.e., the adoption of Cloud technology). In essence, it expands on Ajzen's (1985, 1991) Theory of Planned Behavior in three significant ways. First, our model considers *intention to trust* – rather than *behavioral intention* – as the antecedent to *behavioral outcome*. That is, we focus on the concept of intention to trust as affecting the ultimate behavioral outcome, rather than an implicitly or explicitly expressed behavioral intention (Gefen et al., 2003). Second, our model proposes that, in addition to *subjective norm*, *attitude*, and *perceived behavioral control* (the prime antecedents of behavioral intention in the traditional TPB model), *perceived risk* and *knowledge* are also hypothesized as being antecedents of *intention to trust* (which, in turn, directly affects behavioral outcome). Finally, our model takes into account not only the direct effects of *knowledge*, *attitude* and *perceived behavioral control* on both *intention to trust* and, ultimately, *behavioral outcome*, it also takes into account the moderating effects of *perceived risk* and *subjective norm* on these antecedent factors vis-à-vis *intention to trust* (and, hence, behavioral outcome).

Our research framework, illustrated in Fig. 1, examines causal effects among seven (7) variables based on twelve (12) hypotheses. This framework consists of five predictor variables (*knowledge*, *attitude*, *perceived behavioral control*, *perceived risk* and *subjective norm*) underlying the user's intention to trust. These are considered essential in decision-making (Kamal et al., 2015). *Perceived risk* and *subjective norm* also function as moderator variables, and are hypothesized to influence the three predictor variables where trust intention is the dependent variable. We aim to examine the *moderating* effects of peer influence (as represented by the subjective norm) and the influence of users' perceived risk in their trust intention to adopt the Cloud. Further, we consider behavioral outcome (i.e., actual or

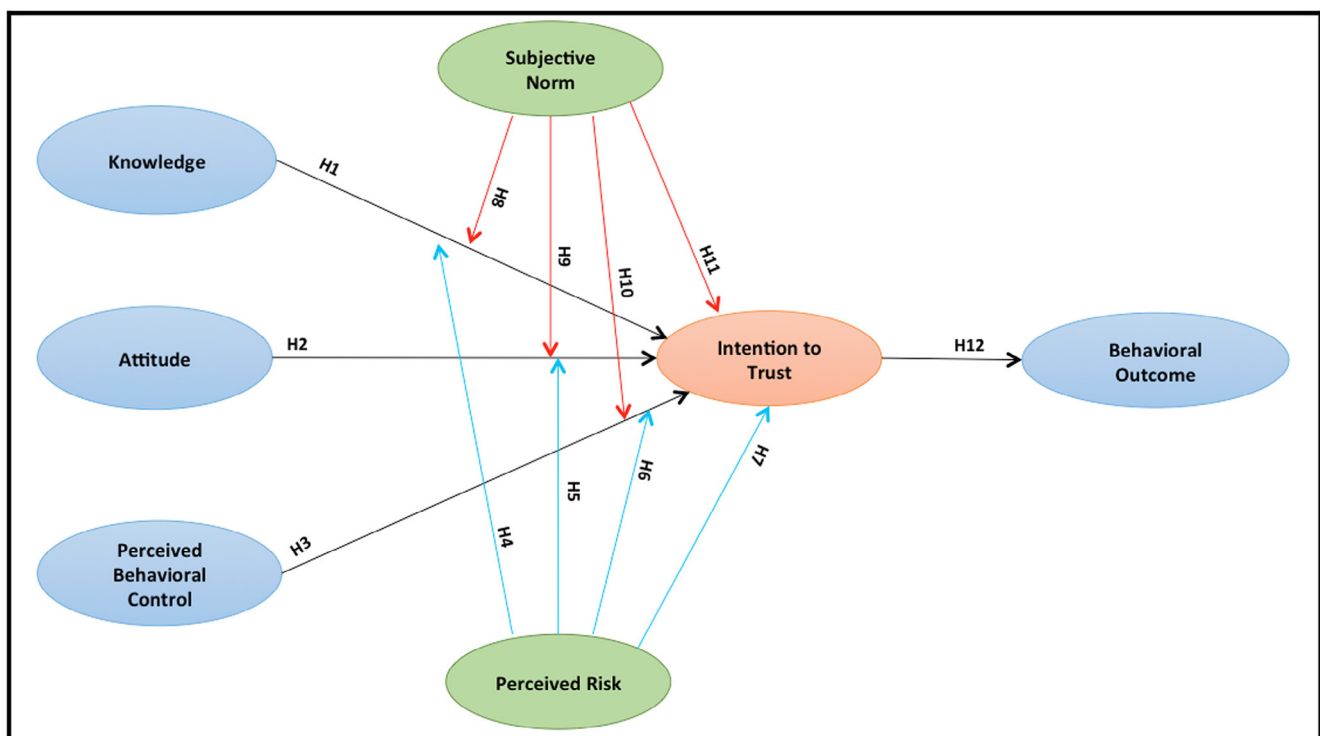


Fig. 1 – Research framework.



attempted adoption) as an additional dependent variable, which may be influenced by some or all of the foregoing variables. The development of each of our hypotheses is discussed and described in the following paragraphs.

### 3.1. Intention to trust

Ajzen (1991) stated that intention to perform an action (i.e., to trust) is an important variable that can affect behavioral outcome in specific contexts (e.g., to adopt technology). Intention to trust can be defined as one's reasonable expectation that another party possesses the characteristics of trustworthiness, which in turn, guides decision-making and action. In this sense, intention to trust is the product of cognitive processes, and the result of individual judgments or cognitive assessments that ascribe benefits and/or costs to potential outcomes of a particular behavior – in this case, the decision to rely on an entity (e.g., a Cloud service provider) to do what it says it will do (i.e., to provide current, convenient, accurate and secure data curation services). Gefen et al. (2003) emphasized that, particularly in e-Commerce, the degree of trust one party has towards the other party weighs heavily in his/her decision to transact business with that party. Intention to trust is therefore an important consideration in the decision to adopt a technology. Indeed, Gefen et al. (2003) suggested that trust should be considered a third factor – alongside perceived usefulness and perceived ease of use – that has direct effect on the adoption of technology.

Trust – or, more precisely for our model, the intention to trust – speaks to an affective or emotional process that results in one's feeling secure when relying on another, based on a person's gut feeling and faith (Komiak and Benbasat, 2004, 2006). Komiak and Benbasat (2006) suggested that the perceived personalization and familiarity of a system's interface can create this sense of security for its users, and thus will influence users' intention to rely on the technology. This type of “evaluative affect” influences the user's attitude toward “relying on” a technology (Komiak and Benbasat, 2006, p. 944), and thus influences his/her intention to trust the technology. The user's intention to trust a technology informs his/her adoption decision-making process.

In considering whether to adopt cloud technology, the decision-making of IT management can likewise be influenced and affected by an intention to trust – i.e., to trust the Cloud infrastructure, or a particular provider of Cloud services.

### 3.2. Knowledge

Knowledge represents a set of facts, information, skills or experiences a person acquires over time. Knowledge acquisition is an aggregative process of collecting, managing, storing, transferring and applying these sets of facts, skillsets and experiences in order to make reasonable decisions, obtain effective results and reach optimal performance in a situation. Raju et al. (1995) described knowledge as subjective and tacit; knowledge requires an individual's subjective “feeling and thinking agent” to synthesize the informational object(s) for better decision-making. For example, in an organizational context, the knowledge of an organization with respect to a particular

decision is frequently influenced and informed by – or supplemented with – social, environmental, and economic concerns. Edmondson et al. (2003) further suggested that knowledge is not only tacit, but codified. Tacit knowledge is normally context specific, and refers to the state of the world (i.e., “know-what”) and competence (i.e., “know-how”) (p. 199). Codified knowledge, on the other hand, is transferrable in formal, symbolic language, which includes experience and information, as well as instruction. Knowledge – whether tacit or codified – is intangible, and improves our ability to make appropriate decisions, especially in organizations (Courtney, 2001). Technology users and decision-makers must have knowledge and understand innovation well in order to adopt technology. Edmondson et al. (2003) thus suggested that knowledge has a direct effect on performance improvement.

In the Cloud environment, where access to information is highly distributed and ubiquitous, the more knowledge one has concerning the Cloud, the better one will understand the particular way(s) in which the Cloud poses a security threat to information. Thus, knowledge of the Cloud should allow a user to make a better-informed decision to trust and to adopt Cloud technology. Accordingly, we posit that:

**Hypothesis 1 (H1).** *Corporate users' knowledge about Cloud technology has a direct effect on their intention to trust Cloud services.*

Also, the effect of users' knowledge on intention to trust can be modified by other factors, which are discussed in Sections 3.5 and 3.6 below.

### 3.3. Attitude

Attitude refers to an expression of a person in favor (or disfavor) of an object or a concept in a situation (Ajzen and Fishbein, 1980). Attitude is an affective or evaluative belief toward a certain behavior, and has long been shown to influence behavioral intentions (Schultze, 2000). Taylor and Todd (1995) empirically found that attitude can directly influence a user's intention to adopt technology. Likewise, Chau and Hu (2001) also noted that attitude is an important factor influencing a professional user's decisions. Paul A. Pavlou (2003) stated that attitude is a function of behavioral beliefs, which are derived from the likelihood of certain outcomes that result from the behavior and evaluation of those outcomes. Attitude is also a function of trust: a trustor's intention to trust will be influenced by his or her assessment as to whether or not the trustee will do what was promised. This assessment informs the trustor's expectations, and also influences the trustor's attitude toward consequences if the trustee does not do what s/he has promised (Jones, 1996; Komiak and Benbasat, 2006).

In the context of corporate users' adoption of knowledge management systems (KMS), Lai (2009) found that the attitude of corporate employees can often be influenced by self-produced factors (i.e., computer self-efficacy), as well as perceived benefits (i.e., reward) and cost stimuli (i.e., perceived power security). Moreover, if corporate users have a positive attitude, the perceived usefulness and perceived ease of use of the KMS will increase and directly influence their satisfaction. Their resultant level of satisfaction will then indirectly influence their intention to adopt KMS (Lai, 2009).

The role of a user's attitude can vary across different technology acceptance models, and may change with the evolution of a particular model. For example, Venkatesh et al. (2003) did not include attitude, self-efficacy or anxiety as direct determinants of behavioral intention in their unified technology acceptance model. However, Venkatesh and Goyal (2010) subsequently categorized perceived usefulness as a factor influencing attitude in their expectation-disconfirmation theory of technology adoption. In Cloud adoption, we consequently hypothesize that a corporate user who has a generally positive attitude toward managing the corporation's data in the Cloud environment, and feels comfortable doing so, will be more likely to trust and to adopt the Cloud. Accordingly, we frame our second hypothesis as follows:

**Hypothesis 2 (H2).** *Corporate users' attitude has a direct effect on their trust intention to adopt Cloud services.*

In addition, attitude can be influenced by other moderating variables in the model, and will be further discussed in Sections 3.5 and 3.6 below.

### 3.4. Perceived behavioral control

Perceived behavioral control (PBC) refers to one's perception as to his/her ability to perform a particular behavior (Ajzen, 1991). PBC not only directly affects the behavioral intention, but also directly affects behavioral outcome (whether attempted or actual). Ajzen (1991, 2002) suggested that PBC is essentially different from Bandura's (1991) perceived self-efficacy, which is the belief system one has over their own capabilities to execute actions. However, Taylor and Todd (1995) suggested that PBC is significantly influenced by self-efficacy (i.e., the individual's belief that they have the ability to perform), and facilitating conditions (i.e., access to the resources and facilities required). Dinev and Hart (2004) further suggested that PBC is one of the antecedents to individuals' privacy concerns. Paul A. Pavlou and Fygenson (2006) also asserted that PBC, as a two-dimensional construct of self-efficacy and controllability, is a key determinant in the external control beliefs for purchasing intention in e-Commerce transactions. Elie-Dit-Cosaque et al. (2011) also suggested that PBC is influenced by internal forces (i.e., personality, character traits), personal innovativeness, external forces (i.e., work environment), and social factors (i.e., management support). In sum, understanding the various determinants for perceived behavioral control is important in assessing organizational IT users' intention to trust and adopt IT systems. Based on the foregoing discussion, we hypothesize that:

**Hypothesis 3 (H3).** *Corporate users' perceived behavior control over the corporate data has a direct effect on their intention to trust Cloud services.*

As with both of the foregoing predictor variables (knowledge and attitude), there would be a moderating influence on PBC from perceived risk and subjective norm as hypothesized in Sections 3.5 and 3.6.

### 3.5. Perceived risk

Perceived risk may be thought of as an individual's subjective belief and expectation of potential harm (i.e., loss or risk), resulting from a particular situation or a set of circumstances, and thus influence the process of decision-making. When this expectation exceeds the individual's risk tolerance threshold, perceived risk would negatively affect a user's intention to purchase products or services (Venkatesh and Goyal, 2010). However, perceived risk is likely to be shaped or influenced by cultural background and experience (Keil et al., 2000). For example, an individual who has been in a car accident will likely consider riding in a car to be more dangerous than someone who has not been in an accident.

Ring and van de Ven (1994) considered the risks involved in e-Commerce transactions to be either technology-driven (i.e., risk from the underlying infrastructure) or relationship-driven (i.e., risk from a specific trading partner). Paul A. Pavlou (2003) suggested that perceived risk is an independent variable that influences shoppers' behavioral intention to transact. Dinev and Hart (2006) also confirmed that perceived risk can influence personal decision and willingness to disclose information online. Thus, perceived risk negatively affect the behavioral intention to transact in e-Commerce environment.

Keil et al. (2000) noted that IT managers always face a variety of risks during IT projects; they must decide whether to accept these risks or not. In particular, their risk propensity can and does affect their intention to undertake or to continue an IT project. Extending this concept to corporate users, these risks may include having their organizational data being comingled with another companies' data, or their data being accidentally disclosed to the public through the Cloud. Their overarching concern is the risk of losing control of their data (Yeh and Chang, 2007). Liu et al. (2008) discussed this problem and noted that, while the adoption of Cloud and other virtualization technology has significantly enhanced IT usage and performance within organizations, it has also clearly led to an increase in organizational decision makers'/ IT managers' concerns about the intangible, undefined boundaries of the Cloud and the increase in risk to data security and integrity (Liu et al., 2008). Since storage in the Cloud has been rapidly developed and deployed at the global level (Ward and Sipior, 2010), the user's sense of uncertainty as to the extent of the risk has increased drastically. It is interesting to note that Liu et al. (2008) examined organizational intent to adopt virtual technologies not through a classic TAM model, but instead through the blended lens of three business-management theoretical frameworks.<sup>1</sup> More specifically, following TPB, Liu et al. (2008) mapped users' attitude to *net perceived benefits*, subjective norms to *external influences*, and perceived behavior control to *organizational capability*. This analysis led them to the conclusion that IT managers will calculate uncertainty, evaluate institutional pressure, and assess not only the controllability, but also risk in situations of uncertainty.

Based on the foregoing, there is clearly an expectation of risk-taking among corporate users and decision makers that

<sup>1</sup> Specifically, transaction cost theory, institutional theory, and organizational-capability-based theory.

informs and influences their decision to adopt Cloud or other virtualization technologies. In our study, we specifically hypothesize that perceived risk, as a moderating factor, can influence corporate users' knowledge, attitude and behavioral control toward Cloud computing. That is, perceived risk will moderate the relationships between independent variables (i.e., perceived behavioral control, attitude, and knowledge) and the dependent/outcome variable (i.e., trust intention as behavioral outcome). Specifically, the relationship between and among these variables will be weakened when perceived risk is present. In other words, perceived risk may minimize the effect on knowledge, attitude, and perceived behavioral control. To examine this in more detail, we propose the following three hypotheses:

**Hypothesis 4 (H4).** *Corporate users' perceived risk of Cloud computing moderates the influence of knowledge on trust intention with respect to adopting Cloud services.*

**Hypothesis 5 (H5).** *Corporate users' perceived risk of Cloud computing moderates the influence of attitude on trust intention with respect to adopting Cloud services.*

**Hypothesis 6 (H6).** *Corporate users' perceived risk of Cloud computing moderates the influence of perceived behavioral control on trust intention with respect to adopting Cloud services.*

In addition to the above hypothesized moderating effects, corporate users' perceived risk can also be a predictor variable, directly affecting a user's trust intention to adopt Cloud technology. Thus, we raise our hypothesis as follows:

**Hypothesis 7 (H7).** *Corporate users' perceived risk has a direct effect on their trust intention with respect to adopting such technology or services.*

### 3.6. Subjective norm

Subjective norm can be defined as a person's subjective belief that most of his/her "important others" think that s/he should (or should not) behave in a specific way. In other words, subjective norms can influence a user's intention and behavioral outcomes (Ajzen, 1985, 1991). Consequently, subjective norms are an important determinant of intention and behavior to adopt technology. For example, studies by Venkatesh and Davis (2000) and Legris et al. (2003) suggest that subjective norms have a moderating effect on users' perceived risk and usage intention, which is also applicable in the realm of Cloud computing.

Empirical tests of social influence or subjective norms on attitudes toward IT have produced mixed results. It is important to note that subjective norms had not been introduced as an influencing factor in the original iteration of TAM, but has been included subsequently. Davis (1989) believed that subjective norms had a negligible effect on behavioral intention. However, Taylor and Todd (1995) took a different view. In comparing TPB and TAM, Taylor and Todd (1995) found that subjective norms are influenced by peers and superiors (e.g., important others and friends). Venkatesh and Davis (2000) extended TAM to include subjective norms, which they characterized as a social influence process. Venkatesh and

Morris (2000) further identified gender differences in how social influence affects an individual's technology adoption behavior. Men's technology adoption behaviors tend to be influenced by their perception of usefulness, while women were more strongly influenced by their perceptions of ease of use and subjective norms. Venkatesh et al. (2003) further confirmed the role of social influence as subjective norms in their unified theory of TAM. When exploring privacy concerns, Dinev and Hart (2005) suggested that social awareness indirectly influences information users' decision to make transactions online. Schepers and Wetzels (2007) found that subjective norms have a significant influence on perceived usefulness and behavioral intention to use. They concluded that subjective norms are more salient during early adoption or diffusion of technology, given the influence of subjective norms on users' knowledge, beliefs, attitudes, etc. in the first instance (Schepers and Wetzels, 2007).

Consequently, subjective norms can essentially be considered social norms (i.e., social pressure imposed by friends, family and relatives) (Lewis et al., 2003; Taylor and Todd, 1995). Lewis et al. (2003) extended this basic definition to include individual factors (i.e., computer self-efficacy, personal innovativeness with technology), institutional factors (i.e., top management commitment, local management commitment), and social factors (i.e., social norms from the departmental peers, informal circles, professional peers, supervisor and senior leaders). In the context of Cloud technology adoption, we thus define subjective norms as perceived social influences or pressure, determined by the total set of accessible normative beliefs about the expectations of important others. In essence, subjective norms can moderate the relationship between knowledge, attitude, and perceived behavioral control. Specifically, subjective norms may minimize the effect on knowledge, attitude, and perceived behavioral control. To further explore this, we offer the following hypotheses:

**Hypothesis 8 (H8).** *Subjective norms moderate the influence of corporate users' knowledge on trust intention with respect to the adoption of Cloud technologies or services.*

**Hypothesis 9 (H9).** *Subjective norms moderate the influence of corporate users' attitude on trust intention with respect to the adoption of Cloud technologies or services.*

**Hypothesis 10 (H10).** *Subjective norms moderate the influence of corporate users' behavioral control on trust intention with respect to the adoption of Cloud technologies or services.*

Ifinedo (2012) empirically demonstrated that subjective norms have a positive effect on behavioral intention. We thus suggest that subjective norms can also be a predictor variable, and hypothesize as follows:

**Hypothesis 11 (H11).** *Subjective norms have a direct effect on corporate users' trust intention with respect to adopting Cloud services.*

### 3.7. Behavioral outcome

Behavior is usually related to outcomes at the level of aggregated consequences rather than momentary effects (Ajzen, 1985). Projected or anticipated consequences could affect an



**Table 1 – Construct Reliability and Validity.**

|                         | Cronbach's Alpha | rho_A | Composite Reliability | Average Variance Extracted (AVE) |
|-------------------------|------------------|-------|-----------------------|----------------------------------|
| Attitude (A)            | 0.946            | 0.946 | 0.959                 | 0.823                            |
| Behavioral Outcome (BO) | 0.939            | 0.941 | 0.961                 | 0.892                            |
| Moderating Effect 1     | 1.000            | 1.000 | 1.000                 | 1.000                            |
| Moderating Effect 2     | 1.000            | 1.000 | 1.000                 | 1.000                            |
| Moderating Effect 3     | 1.000            | 1.000 | 1.000                 | 1.000                            |
| Moderating Effect 4     | 1.000            | 1.000 | 1.000                 | 1.000                            |
| Moderating Effect 5     | 1.000            | 1.000 | 1.000                 | 1.000                            |
| Moderating Effect 6     | 1.000            | 1.000 | 1.000                 | 1.000                            |
| PBC                     | 0.843            | 0.854 | 0.905                 | 0.761                            |
| Perceived Risk (PR)     | 0.940            | 0.946 | 0.957                 | 0.848                            |
| Subjective Norm (SN)    | 0.818            | 0.826 | 0.880                 | 0.647                            |
| Trust Intention (TI)    | 0.941            | 0.941 | 0.955                 | 0.809                            |
| Knowledge (KN)          | 0.868            | 0.952 | 0.901                 | 0.696                            |

individual's behavior through the influence of thought and belief; thus, positive or negative reinforcement of someone's beliefs can influence their behavior. Ajzen (1985, 1991) asserted that an individual's expectations and assumptions concerning the consequences of a particular behavior is a significant factor in his/her decisions and actions. People in general are more likely to undertake behaviors they believe will result in a favorable or desirable outcome than those for which they anticipate negative consequences. Compeau and Higgins (1995) further emphasized that behavior is an observable response or outcome manifested in a given situation with respect to a specific targeted person. In fact, IT users' attitudes are closely related to their actual use of the system (He and King, 2008), and usage (as the desired behavioral outcome) is the means by which the full potential of the system can be realized (Gefen et al., 2003; Lai, 2009; Liu et al., 2008; Oliveira et al., 2014; Straub et al., 1997; Taylor and Todd, 1995; Venkatesh and Davis, 2000; Venkatesh and Goyal, 2010; Venkatesh and Morris, 2000; Venkatesh et al., 2003). In this study, we hypothesize that trust intention can be a predictor variable, influencing the corporate user's behavioral outcome (aggregated consequence) as a dependent variable. Specifically, we frame our hypothesis as follows:

**Hypothesis 12 (H12).** *Corporate users' trust intention has a direct effect on their intention to adopt Cloud services.*

## 4. Data collection

The data collection and validation are described and discussed in this section. We first outline the data collection process and procedures, as well as the demographics of the target population. We further discuss the reliability and validity of the resulting dataset.

### 4.1. Convenience sampling

Survey research was conducted during September 2013 through February 2014. We employed a convenience sampling technique, distributing survey questionnaires to our target population either online or through personal contact. Our target population consisted of senior management as well as systems

managers, staff in the information systems area, as well as managers and others involved in decision-making in both public and private sectors.

Originally, we sampled survey data from 203 respondents, geographically distributed across several different countries. After removing incomplete data, we derived a total of 170 viable surveys. We further removed data where the respondent indicated s/he was not aware of Cloud computing. Thus, our final dataset consisted of data from a total of 153 respondents – 113 males and 40 females. Of these, 29% are employed as IT Managing Director, 18% are employed as managers, 22% are employed as IT professionals, 5% are employed as programmers and 24% are employed in another profession (as in the area of finance and computer technician). 82% of the respondents had more than five (5) years of IT experience, and only 16% had less than five (5) years of IT experience. 93.5% of the respondents have at least a Bachelor's degree (specifically, 44% have a Bachelor's degree, and a further 50% have a Master's Degree or higher). The age of the respondents was in the range of 31–50 years of age.

### 4.2. Reliability and validation of the research model

The questions in the survey instrument called for responses on a 7-point Likert scale ("strongly disagree" to "strongly agree"). Due to the small sample size, we employed partial least square structural equation modeling (PLS-SEM) approach to assess the latent variables in the research model (Hair et al., 2012). In particular, SmartPLS 3.2.6 was used to run the multivariate analysis. We first conducted a reliability test to validate the constructs and to assess the internal consistency of the survey instrument. We derived Cronbach's Alpha values for seven factors in the range of 0.9 (very close to 1) (Table 1). The convergent validity has been established with the average variance extracted (AVE) above 0.7, and the composite reliability (CR) at the range of 0.9 (above the acceptable 0.7) (Table 1). We are thus confident that the survey instrument used is highly reliable.

Then, the factor analysis was performed. The results suggest that our proposed seven-factor-model<sup>2</sup> is appropriate given the

<sup>2</sup> The seven (7) factors include Subjective norm, Attitude, Knowledge, Perceived Behavioral Control, Perceived Risk, Trust intention, and Behavioral Outcome.



**Table 2 – Factor loadings.**

|       | Constructs                                                                                                                             | Factor Loadings |
|-------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| A1    | Using cloud storage is a right and helpful idea.                                                                                       | 0.886           |
| A2    | Using cloud service is an acceptable solution to my corporate data storage.                                                            | 0.868           |
| A3    | Using cloud storage is a good idea.                                                                                                    | 0.955           |
| A4    | I am excited about the idea of using the cloud storage.                                                                                | 0.889           |
| A5    | Using cloud storage is a positive idea.                                                                                                | 0.935           |
| PR7   | I believe that my corporate data stored on, and managed by, this cloud storage service provider is secure.                             | 0.940           |
| PR8   | I am comfortable that the service provider of this cloud storage solution will not use unsuitable methods to collect my personal data. | 0.889           |
| PR10  | I perceive that my corporate data stored on, and managed by, this cloud storage service provided is well protected.                    | 0.943           |
| PR 11 | I believe the service provider of this cloud storage solution will perform due diligence, and secure our corporate data.               | 0.909           |
| KN12  | I have sufficient knowledge about the cloud storage security.                                                                          | 0.865           |
| KN13  | I possess enough knowledge to use and work with cloud storage.                                                                         | 0.843           |
| KN14  | I have sufficient experience in knowing the security of the cloud storage services.                                                    | 0.833           |
| KN15  | I am confident that the service provider has sufficient and knowledgeable technical personnel to manage and secure the cloud storage.  | 0.792           |
| PBC17 | I am certain that the personal information I provide to the services provider is secure.                                               | 0.915           |
| PBC18 | I feel I have sufficient control over the methods used by my service provider in collecting my personal data.                          | 0.855           |
| PBC21 | Using the cloud storage service is under my control.                                                                                   | 0.845           |
| SN23  | Most people who are important to me think it is a good idea to use cloud storage.                                                      | 0.807           |
| SN24  | The advertisement of the services provider influence me in my decision whether to use cloud storage.                                   | 0.737           |
| SN25  | The competitiveness in my industry influences me in my decision whether to use cloud storage.                                          | 0.813           |
| SN26  | Most people who are important to me would use cloud storage.                                                                           | 0.856           |
| TI27  | For me, using the cloud storage in the next six months is important.                                                                   | 0.906           |
| TI28  | I plan to use cloud storage in the next six months.                                                                                    | 0.935           |
| TI29  | I anticipate I will use cloud storage in the next six months.                                                                          | 0.927           |
| TI30  | The Platform as a Service (PaaS) of this service provider is dependable and honest in providing secure cloud storage services.         | 0.873           |
| TI31  | The services provider is competent and trustworthy in handling and securing my data.                                                   | 0.853           |
| BO33  | Is my company currently likely to adopt and use the cloud storage technology?                                                          | 0.920           |
| BO34  | Is my company likely to continuously use the cloud storage technology in the next three years?                                         | 0.970           |
| BO37  | Do I expect my company to continuously use the cloud storage technology in the next three years?                                       | 0.943           |

variables measured, and sufficiently minimizes variable overlap. Hair et al. (2014) suggested the standardized outer loading to be above 0.708 so that the indicator's variance, as explained by the square of the standardized indicator's outer loading ( $0.708^2$ ), is above 50% to ensure indicator reliability (p. 103). Table 2 illustrates the outer loadings of all indicators for associated constructs are greater than all the loadings on other constructs. After removing indicators with outer loadings lower than 0.708 to establish indicator reliability, we establish the discriminant validity for all constructs (Table 2). Thus, we are confident that a sufficient amount of the variance of each item is explained collectively by the seven factors selected.

Correlation analysis was performed to establish any correlations between and among the variables. Table 3 indicates strong correlations ( $p < 0.05$ ) among the variables in the model.

Table 4 illustrates the variance inflation factor (VIF) for the predictor variables of our model. The model shows that four (4) VIFs of the moderator variables can be linearly predicted from the others with a substantial degree of accuracy. Thus, there is multicollinearity between and among these four moderator variables.

## 5. Hypothesis testing

We use partial least squares structural equation modeling (PLS-SEM) analysis to identify and explain the causal relationships

between and among the variables. We further assess latent variables that may have moderating effects in addition to direct effect among all the constructs. As an alternative to covariance-based (CB) SEM, PLS-SEM places more emphasizes on prediction with less demands on “data and specification of relationships” (Hair et al., 2012, p. 415). Thus, we deem PLS-SEM to be an effective approach for examining this dataset with small sample size.

### 5.1. Structural modeling with PLS algorithm

The variables in our research model include *knowledge* (as an independent variable), *attitude* (as an independent variable), *perceived behavioral control* (as an independent variable), *subjective norm* (as both an independent as well as a moderator variable), *perceived risk* (as both an independent as well as a moderator variable), *trust intention* (as a dependent variable), and *behavioral outcome* (as a dependent variable) of adopting Cloud technology. The variability between all predictor variables and *trust intention* is  $R^2 = 0.728$ , and the variability between *trust intention* and *behavioral outcome* is  $R^2 = 0.434$ .

The PLS algorithm further identified interaction effects between the moderator variables (*subjective norms* and *perceived risk*) as well as the exogenous latent variables (*knowledge*, *attitude*, and *perceived behavioral control*). The path coefficients with values above 0.10 show statistically significant interaction effects of the moderator variables on the exogenous latent

Table 3 – Interconstruct correlations.<sup>a</sup>

|                     | Attitude | Behavioral Outcome | Moderating Effect 1 | Moderating Effect 2 | Moderating Effect 3 | Moderating Effect 4 | Moderating Effect 5 | Moderating Effect 6 | PBC   | Perceived Risk | Subjective Norm | Trust Intention | Knowledge |
|---------------------|----------|--------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|-------|----------------|-----------------|-----------------|-----------|
| Attitude            | 1.000    |                    |                     |                     |                     |                     |                     |                     |       |                |                 |                 |           |
| Behavioral Outcome  | 0.498    | 1.000              |                     |                     |                     |                     |                     |                     |       |                |                 |                 |           |
| Moderating Effect 1 | –0.235   | –0.225             | 1.000               |                     |                     |                     |                     |                     |       |                |                 |                 |           |
| Moderating Effect 2 | –0.480   | –0.255             | 0.717               | 1.000               |                     |                     |                     |                     |       |                |                 |                 |           |
| Moderating Effect 3 | –0.298   | –0.235             | 0.566               | 0.761               | 1.000               |                     |                     |                     |       |                |                 |                 |           |
| Moderating Effect 4 | –0.213   | –0.089             | 0.782               | 0.637               | 0.537               | 1.000               |                     |                     |       |                |                 |                 |           |
| Moderating Effect 5 | –0.584   | –0.249             | 0.522               | 0.835               | 0.663               | 0.599               | 1.000               |                     |       |                |                 |                 |           |
| Moderating Effect 6 | –0.370   | –0.170             | 0.373               | 0.597               | 0.751               | 0.587               | 0.770               | 1.000               |       |                |                 |                 |           |
| PBC                 | 0.671    | 0.486              | –0.151              | –0.267              | –0.353              | –0.108              | –0.358              | –0.356              | 1.000 |                |                 |                 |           |
| Perceived Risk      | 0.761    | 0.532              | –0.197              | –0.379              | –0.320              | –0.153              | –0.481              | –0.432              | 0.812 | 1.000          |                 |                 |           |
| Subjective Norm     | 0.698    | 0.484              | –0.259              | –0.394              | –0.373              | –0.233              | –0.413              | –0.322              | 0.704 | 0.676          | 1.000           |                 |           |
| Trust Intention     | 0.724    | 0.659              | –0.264              | –0.373              | –0.378              | –0.142              | –0.404              | –0.328              | 0.732 | 0.754          | 0.749           | 1.000           |           |
| Knowledge           | 0.488    | 0.424              | –0.384              | –0.223              | –0.160              | –0.291              | –0.186              | –0.098              | 0.594 | 0.529          | 0.646           | 0.576           | 1.000     |

<sup>a</sup>All correlations are significant at the  $p < 0.05$  level.

Table 4 – Inner Variance Inflation Factor (VIF) Values.

|                           | Trust Intention |
|---------------------------|-----------------|
| Knowledge (H1)            | 2.653           |
| Attitude (H2)             | 3.752           |
| PBC (H3)                  | 4.274           |
| Moderating Effect 4 (H4)  | 4.389           |
| Moderating Effect 5 (H5)  | 7.556           |
| Moderating Effect 6 (H6)  | 6.545           |
| Perceived Risk (H7)       | 4.663           |
| Moderating Effect 1 (H8)  | 4.729           |
| Moderating Effect 2 (H9)  | 8.539           |
| Moderating Effect 3 (H10) | 5.747           |
| Subjective Norm (H11)     | 3.064           |

Four variance inflation factors (VIFs) of the moderator variables (H5, H6, H9, H10) can be linearly predicted from the others with a substantial degree of accuracy. Thus, multicollinearity exists between and among these four moderator variables.

variables. The results demonstrate that the predictor variables *attitude* ( $\beta = 0.272$ ), *perceived risk* ( $\beta = 0.264$ ), and *subjective norm* ( $\beta = 0.275$ ) have a direct effect on corporate users' trust intention. However, the predictor variables *knowledge* ( $\beta = 0.071$ ) and *perceived behavioral control* ( $\beta = 0.071$ ) do not have a direct effect on users' trust intention. Nonetheless, *trust intention* has a direct effect on users' *behavioral outcome* ( $\beta = 0.659$ ) in Cloud technology adoption.

Moreover, the moderator variable *perceived risk* influences the relationship between *knowledge* and *trust intention* ( $\beta = 0.174$ ). However, there is no statistically significant moderating effect of *perceived risk* on either the relationship between *attitude* and *trust intention* ( $\beta = 0.018$ ) or the relationship between *perceived behavioral control* and *trust intention* ( $\beta = 0.008$ ).

Likewise, the moderator variable *subjective norm* influences the relationship between *knowledge* and *trust intention* ( $\beta = -0.147$ ), and the relationship between *perceived behavioral control* and *trust intention* ( $\beta = -0.154$ ). However, *subjective norm* did not have a statistically significant moderating effect on the relationship between *attitude* and *trust intention* ( $\beta = 0.098$ ) (Fig. 2).

## 5.2. PLS bootstrapping

We ran a bootstrapping to create randomly drawn observations from the original set of data, and estimate nonparametric confidence intervals. The bootstrap-based inference generated the standardized root mean square residual (SRMR), which is based on transforming both the sample covariance matrix and the predicted covariance matrix into correlation matrices that allow an assessment of the average magnitude in discrepancies between observed and expected correlations as an absolute measure of model fit criterion. Our model derived SRMR = 0.10, which is considered a good fit. Moreover, the bootstrapping also derived the normed fit index (NFI = 0.722), which is between 0 and 1.

Although PLS-SEM generates model fit indices that help judge how well a hypothesized model structure fits the empirical data, Hair, Hult, Ringle, and Sarstedt (2017) suggested that these model fit assessment criteria are often not useful for PLS-SEM. This is because the PLS-SEM focuses on prediction in the model, not on generalization or the ability to predict

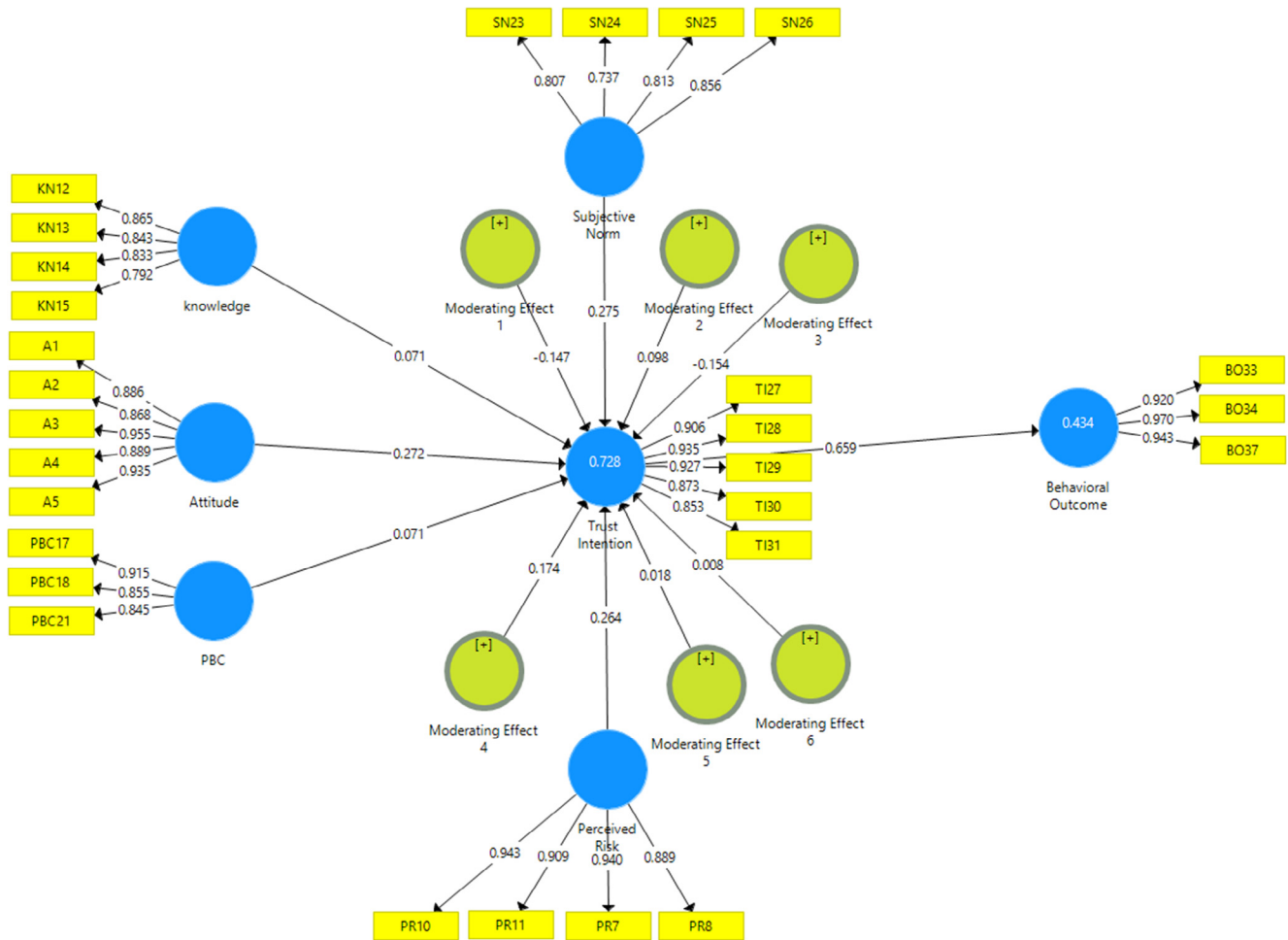


Fig. 2 – PLS with moderation analysis.

sample data as the explanatory model does. Thus, these model fit indices can only be for reference, and offer very little value in model validation.

The bootstrap-based inference tested the significance of estimated path coefficients. The decision as to whether each variable is significant was made based on the critical  $t$  and  $p$  values with one-tailed or two-tailed tests (Hair et al., 2017; Kock, 2015). The critical values for one-tailed tests we employed are 2.33, 1.65, and 1.28 for significance levels of 1%, 5%, and 10% respectively. Likewise, the critical values for two-tailed tests are 2.57, 1.96, and 1.65 for significance levels of 1%, 5%, and 10% respectively (Hair et al., 2017; Kock, 2015). The PLS bootstrapping results demonstrate that the predictor variables, attitude ( $t = 2.844^{***}$ ), perceived risk ( $t = 2.028^{**}$ ), and subjective norm ( $t = 3.357^{***}$ ) have significant relationships with the dependent variable (trust intention). However, the predictor variables, knowledge ( $t = 0.901$ ), perceived behavioral control ( $t = 0.835$ ) do not have a significant relationship with the dependent variable (trust intention). Moreover, the predictor variable trust intention has a significant causal relationship with the dependent variable behavioral outcome ( $t = 12.272^{***}$ ).

The moderator variable perceived risk has a strong and significant interaction effect on the latent exogenous variables knowledge ( $t = 1.670^*$ ). Unfortunately, the perceived risk does not

intervene with the predictor variable attitude ( $t = 0.186$ ) or perceived behavioral control ( $t = 0.086$ ), meaning this moderator variable does not have an effect on corporate user's trust intention in Cloud technology adoption.

Likewise, the moderator variable subjective norm has a strong and significant interaction with the exogenous latent variables knowledge ( $t = 1.746^*$ ), and perceived behavioral control ( $t = 1.616^*$ ) with one-tailed test as these empirical  $t$  values are larger than the critical  $t$  ( $t = 1.28$  at 10% significance level, respectively) (Hair et al., 2017). However, subjective norm does not moderate or intervene with users' attitude ( $t = 1.065$ ) toward trust intention.

### 5.3. Discussion and summary

Table 5 summarizes the findings of the structural model beta coefficient, and bootstrapping with  $t$  and  $p$  values. Five (5) hypotheses were not supported, and seven (7) hypotheses were supported with beta coefficients above 0.10, and through the significance of  $p$  values.

This research provides a causal model that explains both causation and correlation among latent variables; in that certain variables (i.e., subjective norms and perceived risk) not only have direct effects on the user's trust decision, but also more

**Table 5 – Summary of Structural Model.**

| Relationship                                      | Standardized beta coefficient | t-value | p-value   | Hypotheses    |
|---------------------------------------------------|-------------------------------|---------|-----------|---------------|
| H1: Knowledge → trust intention                   | 0.071                         | 0.901   | $p > 0.1$ | Not supported |
| H2: Attitude → trust intention                    | 0.272                         | 2.844   | ***       | Supported     |
| H3: PBC → trust intention                         | 0.071                         | 0.835   | $p > 0.1$ | Not supported |
| H4: Knowledge * Perceived risk → trust intention  | 0.174                         | 1.670   | *         | Supported     |
| H5: Attitude * Perceived risk → trust intention   | 0.018                         | 0.186   | $p > 0.1$ | Not supported |
| H6: PBC * Perceived risk → trust intention        | 0.008                         | 0.086   | $p > 0.1$ | Not supported |
| H7: Perceived risk → trust intention              | 0.264                         | 2.028   | **        | Supported     |
| H8: Knowledge * Subjective norm → trust intention | −0.147                        | 1.746   | *         | Supported     |
| H9: Attitude * Subjective norm → trust intention  | 0.098                         | 1.065   | $p > 0.1$ | Not supported |
| H10: PBC * Subjective norm → trust intention      | −0.154                        | 1.616   | *         | Supported     |
| H11: Subjective norm → trust intention            | 0.275                         | 3.357   | ***       | Supported     |
| H12: Trust intention → behavioral outcome         | 0.659                         | 12.272  | ***       | Supported     |

\* $p < 0.1$   
\*\* $p < 0.05$   
\*\*\* $p < 0.001$

importantly have a moderating effect on other variables (i.e., *knowledge*, *attitude* and *perceived behavioral control*) toward Cloud technology adoption. The results demonstrate that *perceived risk* and *subjective norm* (as predictor variables) have a positive effect on *trust intention* and *behavioral outcome*. Moreover, *perceived risk* significantly moderates corporate users' *knowledge* about Cloud technology adoption. However, *perceived risk* does not motivate users' *attitude* to adopt the Cloud, but also makes users feel powerless in controlling (i.e., their *perceived behavioral control* over) corporate information and intellectual property because of the fact that in the Cloud environment, the Cloud provider becomes the primary data owner of their corporate proprietary data.

Users' *knowledge* ( $\beta = -0.147$ ) and *perceived behavioral control* ( $\beta = -0.154$ ) toward the Cloud are often negatively influenced by social norms – as noted by the negative value of beta coefficient – when making decisions about adopting Cloud service. While users' *attitude* and excitement toward the Cloud significantly influence their trust intention of Cloud technology, their *attitude* is neither influenced by their *perceived risk* nor by *subjective norms*.

## 6. Contributions to research and practice

This study explores the critical role played by a user's *intention to trust* and to ultimately adopt the Cloud technology. This study demonstrates that *intention to trust* can be an important factor influencing corporate users' decision-making process to adopt the Cloud technology. Our research contributes uniquely to the theoretical framework, in that we have validated a causal model that explains the causal mechanisms specific to Cloud technology adoption. This model can also be extended to examine other related issues such as privacy of personal information in online transactions. From a theoretical perspective, the main contribution of our work is a recomposed TPB, which considers *intention to trust*, rather than *behavioral intention*, as an immediate antecedent of *behavioral outcome*, which is useful in explaining users' ultimate decisions to adopt (or not) a new technology – such as Cloud computing.

This research further contributes to the practitioners' community in both the private and public sectors. Cloud providers

should understand that a corporate user's *attitude* strongly influences their *intention to adopt the Cloud*, but their *attitude* is not influenced by their *knowledge of the Cloud* or by social influences such as marketing. Cloud service providers should thus focus on users' *attitude* – and factors that moderate *attitude* – when advertising their products and services. In particular, manufacturers/ developers of Cloud technologies should address corporate users' risk perceptions of the Cloud, and delegate data control to these corporate users so as to reduce the sense of lack of control. Certain considerations such as branding and marketing (e.g., shaping social norms) that emphasize the usefulness and usability aspect of the Cloud will enhance users' attitudes, while addressing the risk perception by communicating how the Cloud environment is securely managed. These considerations can improve corporate users' *intention to trust* and to adopt the Cloud.

## 7. Limitation and future research

This study has certain considerations and limitations, which we must acknowledge. First, because Cloud technologies were designed specifically to be easy to use and useful, these classic TAM factors – *perceived usefulness* and *perceived ease of use* – are assumed as being constant, and have not been specifically or explicitly incorporated into our analysis.

A second potential limitation of our study is that, while the survey was distributed to a global audience, the study (by design) does not identify participant counts by continent. Third, gender distribution is unequal, which may introduce gender bias. However, our research model and the findings are nonetheless generalizable because data was collected through various channels from human subjects in different companies and government agencies across continents. Fourth, this research is limited (by design) to examine only individual organizational user's cognitive perspectives; it does not consider organizational environment factors, cultural factors, and any organization's actual decision on Cloud technology adoption. Future research is expected to study other variables that may influence trust intention, such as cost and information privacy concern.



## 8. Conclusion

In the business world, information security involves mitigating risks to the organization's information assets. These risks, for the most part, have been perceived as technical issues that most executives rarely consider (Garfinkel, 2012). This research argues that an organization's decision to implement information security in the Cloud environment is not just a technical issue; it is affected by non-technical factors such as corporate users' knowledge, attitude and behavioral control over corporate proprietary information stored in the Cloud. Moreover, corporate users' intention to trust Cloud technology is not only directly affected by their attitude, perceived risk and subjective norms toward the Cloud, but also indirectly influenced by perceived risk associated with their own information, and subjective norms from their social networks. These perceived risks and subjective norms indirectly moderate corporate users' perceived behavioral control, such that they feel that their knowledge of the Cloud is insufficient for effective decision-making vis-à-vis adopting the Cloud, and that they lack control over their own proprietary information.

In order to successfully migrate to the Cloud environment, it is essential for organizations to retain knowledge workers, to comply with a variety of legal requirements, to diligently maintain access control standards, and to align system policies with management policies (Ransbotham and Mitra, 2009). These countermeasures will enhance knowledge workers' perception of information control, reduce perceived risk, enhance confidence and trust intention, while avoiding the consequences of knowledge workers being left with little or no control over organizational information assets. Moreover, the solution to transfer total control of corporate users' own information to these corporate users will foster a desired behavior to defend against opportunistic hackers compromising information, and the Cloud information systems and networks.

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